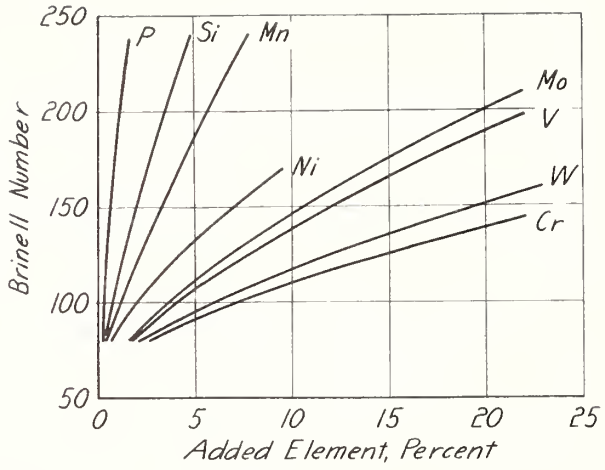


FIGURE 79. — *Probable hardening effect of various elements in pure iron (Bain [661]).*

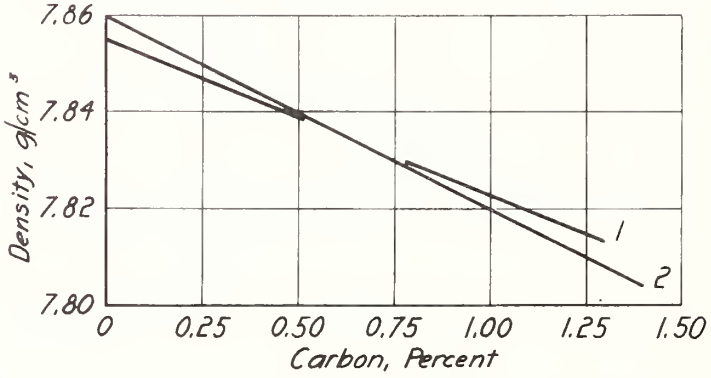


FIGURE 80. — *Densities of hot-rolled and annealed carbon steels (Cross and Hill [662]).*

Curves: 1, hot-rolled; 2, annealed.

The densities are given approximately by the following formulas:

Hot-rolled....density=7.855-0.032 C
Annealed.....density=7.860-0.04 C
(C=carbon content in percent)

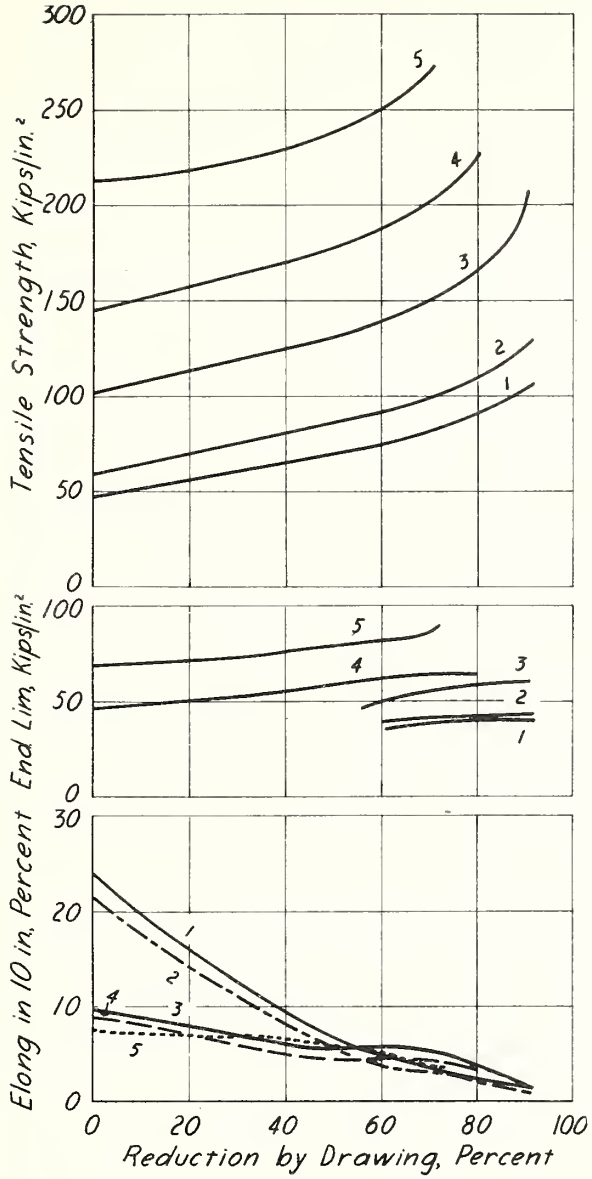


FIGURE 88.—*Tensile properties and endurance limit of cold-drawn steel wire (Godfrey [664]).*

Curves: 1, C 0.05%, Mn 0.11% (drawn from a hot-rolled green rod); 2, C 0.16%, Mn 0.49% (drawn from a hot-rolled green rod); 3, C 0.39%, Mn 0.75% (drawn from a lead-patented rod); 4, C 0.58%, Mn 0.60% (drawn from wire given an intermittent patenting); 5, C 0.90%, Mn 0.35% (drawn from wire given an intermittent patenting).

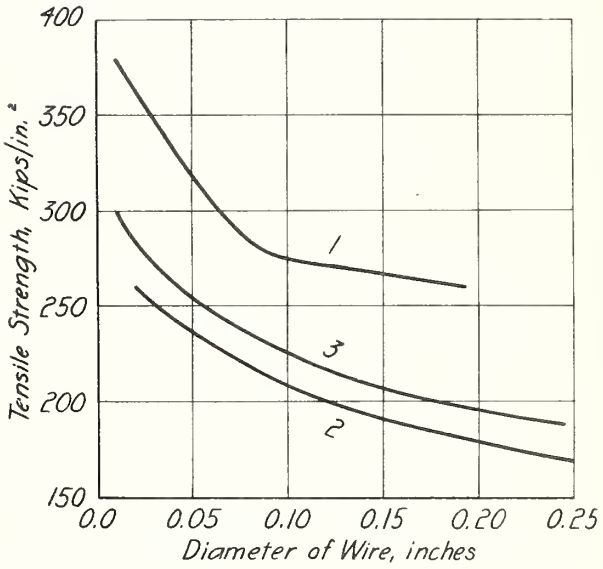


FIGURE 89.—*Tensile strength of steel wire (Eakin [665]).*

Curves: 1, music wire, C 0.70-1.00%, Mn 0.25-0.50%, Si 0.10-0.30%, P 0.025% (max), S 0.025% (max); 2, hard-drawn steel wire, C 0.50-0.70%, Mn 0.70-1.10%, Si 0.10-0.20%, P 0.04% (max), S 0.04% (max); 3, oil-tempered or stainless-steel wire. The oil-tempered wire had a composition similar to that of the hard-drawn wire (2). The stainless-steel wire contained 18% of chromium and 8% of nickel.